

INTRODUCTION TO TIME SERIES ANALYSIS & FORECASTING :

A Python Implementation

I. Intro

1. Definitions:

A time series (ts) = a sequence of observations recorded at regular time intervals, and relative to a **single phenomenon**. By definition, the frequency of observations aka **Periodicity**, is considered to be constant through time and may be:

- *hourly, daily, weekly, monthly, quarterly or annual.*

Also possible to see seconds and minute-wise time series, eg: number of clicks and user visits every minute etc.

In the framework of this presentation, the data to be handled is considered to contain no missing values and no outliers, else, prior cleansing of the data is needed (outside of the scope of this lecture).

As ts at a given time t is denoted X_t with:

X the name of the series;

t time of observation with $t = 1 \dots T$ where 1 is the starting point and T the endpoint or horizon (as also the number of observations)

And X_{T+h} being the forecast of the ts for a given forecast horizon ($X_{T+1}, X_{T+2}, \dots X_{T+n}$)

2. A bit of history

The methodological approach used here is based on the decomposition and reconstruction method in order to forecast a ts. The rationale being that in its compound form, the ts might be a bit complicated to analyze and forecast, but if we managed to decompose it in its individual components, then the task becomes easy.

What are these individual components we're talking about?

Premise: a TS is composed of 4 main components:

- A long-term component, aka a **trend** T_t , depicting the long term or structural Behaviour of a phenomenon. This equates for example to the long business cycles (Juglar – 10+ yrs; Kuznets – 20 years or Kondratieff – 40 to 60 years).
- A lesser long-term component, a **cycle** C_t , depicting the long-term of shorter horizon (3 to 10 years) behaviors of a phenomenon. A Kitchin business cycle (3 to 5 years) is a good example of this.
 - ⇒ In our case, we are going to merge these 2 components as one representing the entirety of the long-term dynamics of a ts. The combined components will be referred to as the trend.
A trend is observed when there is an increasing or decreasing slope observed in the time series.
- A medium-term component, aka a **seasonal component** S_t (or Seasonality), having a somewhat regular behavior and contained within a year. This behavior coincides with fashionable phenomena, cultural or societal customs, or climate related phenomena.

⇒ *Seasonality is observed when there is a distinct repeated pattern observed between regular intervals due to seasonal factors. It could be because of the month of the year, the day of the month, weekdays or even time of the day.*

- A short-term component, aka a **residual component** R_t , represents short-term events that cannot be accounted for by the previous components. . It represents the random effects in a time series, caused by unforeseen rare events such as wars, hurricanes, strikes, by randomness of human actions.

⇒ *The Residual component is the only one that is considered to be present within a series by default*

A time series does not necessarily always exhibit a trend and/or seasonality. One may not have a distinct trend but have a seasonality, just as the opposite can also be true. Hence, a TS may be envisioned as a combination of the trend, seasonality and the residual compounds.

Time series analysis, via Decomposition – Reconstruction involves understanding various aspects about the inherent nature of the series, in order to be better informed to create meaningful and accurate forecasts.

3. Prerequisites on the decomposition methodology

From everything seen previously, it appears that a time series is a given combination of a trend, a seasonality and a residual component.

$$X_t = T_t \blacksquare S_t \blacksquare R_t$$

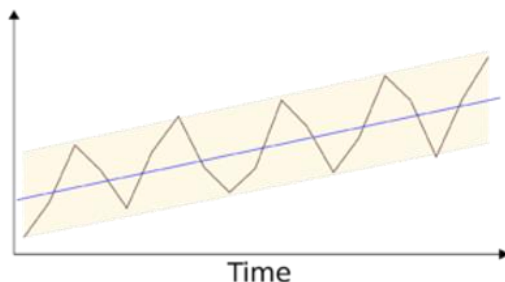
Hence, the decomposition procedure will be dependent upon the specific pattern depicting the way these components interact with each other. There are 2 commonly used types of decomposition patterns or models *i.e.*:

- An additive model, $X_t = T_t + S_t + R_t$, builds on the premise that the components are orthogonal with the seasonality being rigid in amplitude and periods (more on this later).
- A multiplicative model, $X_t = T_t * S_t * R_t$, where all the components interact with each other in a proportional manner.

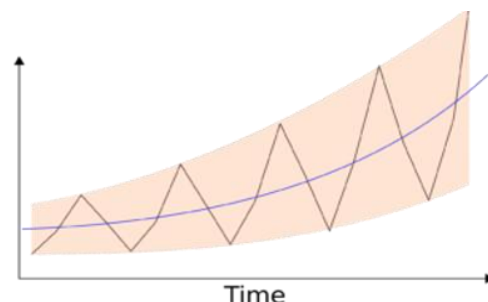
The first focal point of TS analysis is determining how the components interact with each other. This section provides simple methods to do exactly just that.

a. The band procedure.

This first approach to identifying the decomposition pattern relies on a visual examination of the series. The idea is to draw two curves on the graph of the observed series: one connecting the lowest points (minimum) and the other connecting the highest points (maximum).



If the two curves are parallel and form a strip of constant width, we would use the **additive** model



If the two curves are not parallel and if the width of the strip is not constant, it indicates a **multiplicative** model,

b. A tabular method: the Buys-Ballot test

A Buys-Ballot table summarizes data to highlight seasonal variations. Each line represents one period (usually a year) and each column is a season of the period/year (4 quarters, 12 months, etc).

The cells contain the values for all observations made during the various periods. To analyze the data, it is helpful to include the period and seasonal totals, period and seasonal averages and period and seasonal standard deviations as part of the Buys-Ballot table.

Eg: for a quarterly time series available on 3 years, we'd have the following:

Q1	Q2	Q3	Q4	Average ($\bar{p}_i$)	StdDev (σ_{p_i})
Year 1					
Year 2					
Year 3					
Average ($\bar{s}_i$)					
StdDev (σ_{s_i})					

Building from this table, the Buys-Ballot test will evaluate the existence of a linear relationship between the average periodic value and the standard deviation of the periodic values ; *i.e.* $\sigma_{p_i} = \alpha + \beta \bar{p}_i + \varepsilon_i$

The decomposition model is obtained by testing the significance of the β coefficient.

- If the β coefficient is statistically null, then the decomposition model is additive
- Else, if β significantly non null then the decomposition model is multiplicative

In the decomposition and reconstruction procedure for time series analysis, the main steps of the process consist in the following:

- Determine the decomposition model of the series
- Identify presence/absence of the various components (T_t and S_t)
- Extract the various components and forecast them
- Recombine the components and their forecasts in order to obtain the forecasted values of the series